

Thm. Let $f: [a, b] \rightarrow \mathbb{R}$ be a Riemann-integrable, bounded map.

Then, $F: [a, b] \rightarrow \mathbb{R}; x \mapsto \int_a^x f(t) dt$

is continuous uniformly. Moreover, if f is continuous at $x_0 \in [a, b]$, then the derivative

$$F'(x_0) = f(x_0) \text{ exists.}$$

Proof. Let $M > 0$ be a bound for $|f|$. Then, given $x, y \in [a, b]$,

$$|F(x) - F(y)| = \left| \int_y^x f(t) dt \right| \leq \int_y^x |f(t)| dt \leq M \cdot |x - y|.$$

Thus, given $\varepsilon > 0$, we can take $\delta = \frac{\varepsilon}{M}$, to uniformly continuity.

Now, let $x_0 \in [a, b]$ s.t. f is continuous at x_0 .

Given $\varepsilon > 0$, take $\delta > 0$ such that

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

Now, if $0 < |x - x_0| < \delta$, then

$$\begin{aligned} \left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| &= \left| \frac{1}{x - x_0} \cdot \left(\int_{x_0}^x f(t) dt \right) - f(x_0) \right| \\ &= \left| \frac{1}{x - x_0} \cdot \int_{x_0}^x f(t) - f(x_0) dt \right| < \varepsilon. \end{aligned}$$

Thm. If $f: [a, b] \rightarrow \mathbb{R}$ is Riemann-integrable and $F' = f$, then

$$\int_a^b f(t) dt = F(b) - F(a).$$

Proof. Let $P = \{x_0, \dots, x_n\}$ be an arbitrary partition on $[a, b]$.

Then, by the mean-value Theorem, for each $1 \leq i \leq n$,

$$F(x_i) - F(x_{i-1}) = F'(t_i) \cdot (x_i - x_{i-1}) \text{ for some } t_i \in (x_{i-1}, x_i).$$

$$\text{So, } F(b) - F(a) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})) = \sum_{i=1}^n f(t_i) \cdot (x_i - x_{i-1}).$$

$$\text{Now, since } \mathcal{L}(f, P) \leq \sum_{i=1}^n f(t_i) \cdot (x_i - x_{i-1}) \leq \mathcal{U}(f, P),$$

the squeeze Theorem gives the result.